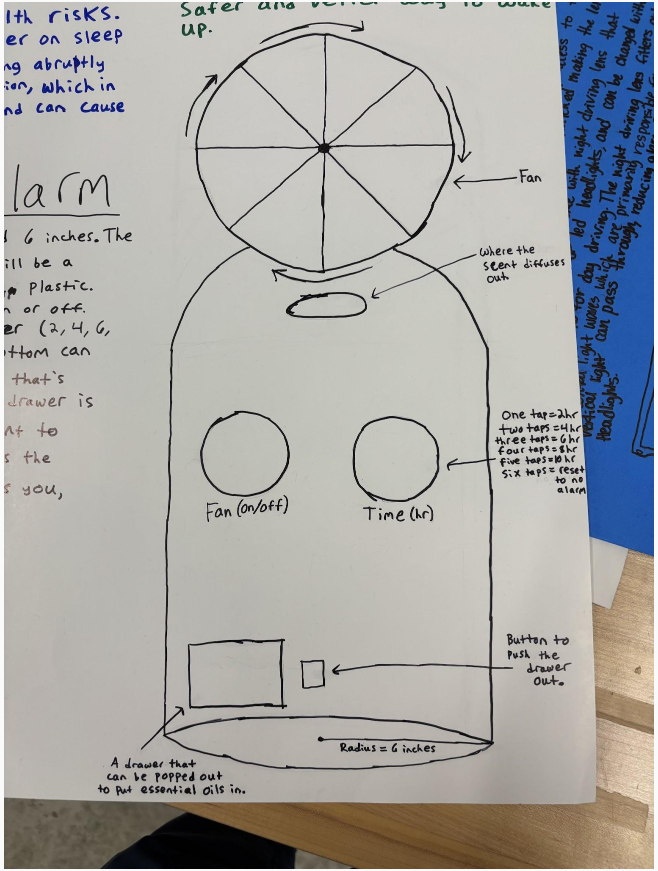
Engineering Project Portfolio

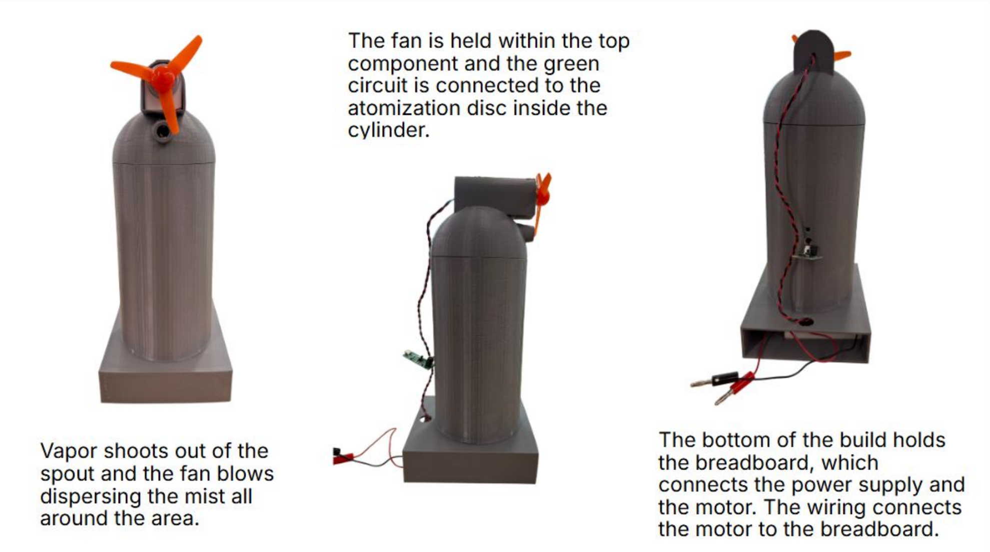
Dylan Miller | Ben Barber Innovation Academy | 2024-2025

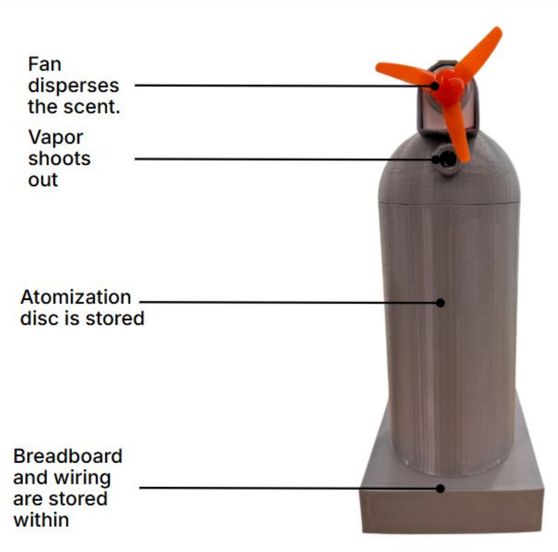
Engineering Design and Development

Developed the first draft of a patent-pending scent-based alarm clock.

Conducted research and personally translated findings into detailed design diagrams: ultimately, guiding development decisions to improve the concept's functionality and effectiveness.





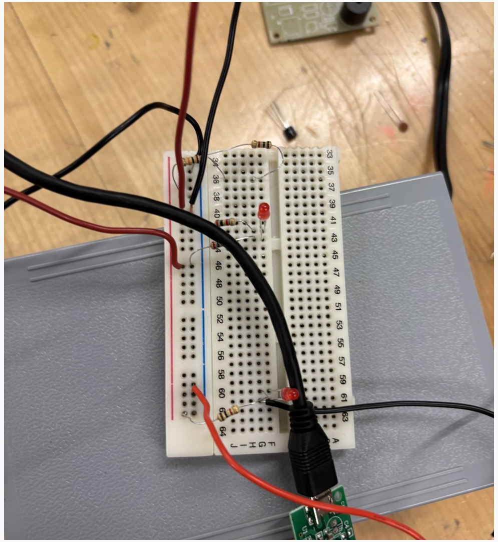


Led the CAD development and physical integration by designing component spacing around the essential-oil reservoir, atomization disk, fan, wiring, and electrical components to produce a functional 3D-printed prototype without requiring a redesign.

Conducted controlled testing across 3+ variables including fan voltage/speed, placement, and scent selection.

Measured mist travel and compared scent intensity to determine conditions that improved scent distribution and overall functionality.

Performed iterative electrical testing that resulted in 2 circuit-board failures after exceeding voltage limits, using the failures to identify operating constraints and adjust component-testing procedures.



Independently modified and soldered wiring to extend leads and integrate system components by incorporating a breadboard into the final assembly while maintaining proper fit within the CAD-designed enclosure.

Project Development Overview

Research

- Researched sleep patterns, waking methods, and existing alarm clock technologies.
- Analyzed existing patents to identify opportunities for innovation.
- Investigated scent effectiveness to guide the alarm clock's design.

Design

- Led the development of the alarm clock concept through sketches, diagrams, and CAD modeling.
- Designed component placement to optimize scent distribution and overall functionality.
- Applied engineering principles to determine the appropriate fan and atomization system.

Development

- Built and assembled the prototype using 3D-printed components and electrical hardware.
- Integrated the atomization disc, fan, circuit board, and scent-delivery system.
- Utilized soldering and breadboard circuitry to develop the working prototype.

Evaluation

- Tested atomization distance and fan speeds at varying voltage levels.
- Evaluated multiple scents to determine their strength and effectiveness.
- Resolved electrical failures and refined the prototype based on testing results.

[View STL File](#)

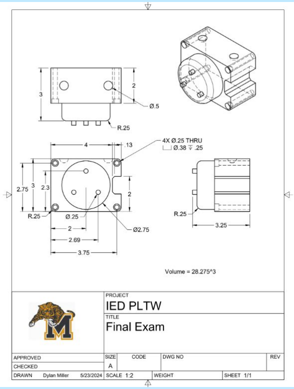
Other Projects

[View STL File](#)

Toy Train

Intro to Engineering Design
Jan 2024 - May 2024

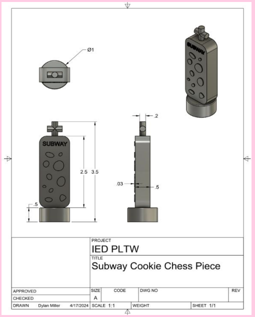
Designed and modeled a two-cart toy train in CAD to meet teacher-provided specifications. Developed a customized Christmas-themed cart featuring integrated trees, presents, and lettering.



Subway Cookie Chess

Intro to Engineering Design
Jan 2024 - May 2024

Given a prompt to combine a chess piece with a fast-food item. I utilized my CAD skills and creative thinking to design a unique piece. The class then combined our 3D-printed creations to assemble a fully functional chess set.



Wind Powered Motor

Computer Integrated Manufacturing
Aug 2024 - Dec 2024

Spent the first half of the semester designing and dimensioning the parts I would manufacture using CAD and industry-standard specifications. Midway through the semester, we transitioned into the lab, where I spent each day learning to operate Haas CNC lathes and mills, eventually applying the skills we developed to manufacture a wind-powered motor.

[View Cams Measurements](#)

Soccer Jersey Charm

Intro to Engineering Design
Jan 2024 - May 2024

Tasked with designing anything of our choice. I created a custom soccer jersey inspired by the current season and my passion for the sport.